

Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

$$\Delta m_{\text{atm}}^2 = 120 m_\nu^2, \quad \Delta m_{\text{sol}}^2 = \frac{11}{3} m_\nu^2$$

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Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

Abstract

Starting from the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (Doc. 339), the measured neutrino mass differences Δm_{atm}^2 and Δm_{sol}^2 are derived from algebraic properties of the four Frobenius orbits in $\mathbb{Z}_{13}$. The atmospheric mass difference $\Delta m_{\text{atm}}^2 = (11^2 - 1)m_\nu^2 = 120 m_\nu^2$ (1.0 % deviation) and the solar $\Delta m_{\text{sol}}^2 = (|\mathbb{Z}_{13}| - 2)/3 \cdot m_\nu^2 = (11/3)m_\nu^2$ (0.46 % deviation) follow from the number 11, the maximum element of the fourth Frobenius orbit $\{7, 8, 11\}$ in $\mathbb{Z}_{13}$. Orbit 3 $\{4, 10, 12\}$ is algebraically self-inverse and carries Majorana character. The coherent venting amplitude $p_{\text{vent}} = 5/24$ is proved algebraically from the branching rule $\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}$ and the index $|A_5 : A_4| = 5$. The SICC venting formula is equivalent to the standard oscillation formula. Status: **[B]** for the group theory; **[K]** for the mass differences.

.1 Starting Point

In Doc. 339 the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ is established [B]. The four three-element orbits of $\mathbb{Z}_{13}$ under the Frobenius $\phi : x \mapsto 3x$ are:

Orbit	Elements	Sum	Product mod 13
1	{1, 3, 9}	13	1
2	{2, 5, 6}	13	8
3	{4, 10, 12}	26	12
4	{7, 8, 11}	26	5

The FFGFT neutrino base mass is speculatively set as $m_\nu = \xi^2/2 \cdot m_e \approx 4.54 \text{ meV}$ [S] (Doc. 047). The present calculations test whether the measured mass differences follow from the Galois structure.

.2 Algebraic Properties of the Orbits [B]

Orbit 4 as a multiple of Orbit 1

$$\{7, 8, 11\} = 7 \cdot \{1, 3, 9\} \pmod{13},$$

since $7 \cdot 1 = 7$, $7 \cdot 3 = 21 \equiv 8$, $7 \cdot 9 = 63 \equiv 11 \pmod{13}$. Since $2^{-1} \equiv 7 \pmod{13}$:

$$\text{Orbit}_4 = 2^{-1} \cdot \text{Orbit}_1 \pmod{13}.$$

Duality Orbit 2 $\leftrightarrow$ Orbit 4

The minimal elements and their multiplicative inverses in $\mathbb{Z}_{13}$:

Orbit	min	$\text{min}^{-1} \pmod{13}$	lies in
1	1	1	Orbit 1
2	2	7	Orbit 4
3	4	10	Orbit 3
4	7	2	Orbit 2

Orbits 2 and 4 are algebraically inverse: $2^{-1} = 7$, $7^{-1} = 2$. Orbit 3 is self-inverse: $4^{-1} = 10$, $10^{-1} = 4$, $12^{-1} = 12 \pmod{13}$. The self-inverse nature of Orbit 3 corresponds algebraically to Majorana character. The sum: $4 + 10 + 12 = 26 = |GF(27)^*|$.

.3 Neutrino Masses from the Orbit Structure [K]

The factor 11

The maximum element of the fourth orbit, $\max(\{7, 8, 11\}) = 11$, follows from:

$$11 = |\mathbb{Z}_{13}| - |\text{Frobenius fixed points in } \mathbb{Z}_{13}| - 1 = 13 - 1 - 1 = 11.$$

This factor determines the mass hierarchy (normal ordering):

State	Galois origin	Mass	Value
ν_1	Orbit 1 (D4, 4D)	m_ν	4.54 meV
ν_2	Orbit 3 (self-inverse, E6, solar)	$\sqrt{14/3} m_\nu$	9.81 meV
ν_3	Orbit 4 (E8, atm., heaviest)	$11 m_\nu$	49.96 meV

The sum $\sum m_i = (1 + \sqrt{14/3} + 11) m_\nu \approx 64.3 \text{ meV}$ is consistent with the Planck-2018 bound $\sum m_i < 120 \text{ meV}$.

Atmospheric mass difference

$$\Delta m_{\text{atm}}^2 = m_{\nu_3}^2 - m_{\nu_1}^2 = (11^2 - 1) m_\nu^2 = 120 m_\nu^2 = 2.476 \times 10^{-3} \text{ eV}^2.$$

Measured: $2.500 \times 10^{-3} \text{ eV}^2$. **Deviation: 1.0 %.**

The factor $120 = |A_5 \times \mathbb{Z}_2| = 11^2 - 1$ is the order of the full icosahedral symmetry group – consistent with the A_5 symmetry of the HLV sector (Doc. 285).

Solar mass difference

$$\Delta m_{\text{sol}}^2 = m_{\nu_2}^2 - m_{\nu_1}^2 = \left(\frac{14}{3} - 1\right) m_\nu^2 = \frac{11}{3} m_\nu^2 = 7.565 \times 10^{-5} \text{ eV}^2.$$

Measured: $7.530 \times 10^{-5} \text{ eV}^2$. **Deviation: 0.46 %.**

The algebraic derivation of $11/3$ [B]:

$$\frac{11}{3} = \frac{|\mathbb{Z}_{13}| - |\text{fixed points of } GF(27)^*|}{|\text{Frobenius order}|} = \frac{13 - 2}{3}.$$

Numerator: $13 - 2 = 11$ (the two fixed points $\{+1, -1\}$ of $GF(27)^*$ subtracted from $|\mathbb{Z}_{13}|$). Denominator: 3 (the order of $\phi : x \mapsto x^3$).

.4 Kappa and the SICC Connection [K]

Relation $\kappa - \Delta m_{\text{atm}}^2$

From Doc. 339 (and Paul Masic's SICC derivation):

$$\kappa = \frac{8}{r_e} \cdot m_e \xi^2 = 6 \xi^2 m_e = 12 m_\nu.$$

It then follows immediately:

$$\Delta m_{\text{atm}}^2 = (11^2 - 1) m_\nu^2 = 120 m_\nu^2 = \frac{5}{6} \kappa^2.$$

The ratio:

$$\frac{\kappa^2}{\Delta m_{\text{atm}}^2} = \frac{144}{120} = \frac{6}{5}.$$

The coherent venting amplitude $p_{\text{vent}} = 5/24$ [B]

If W-venting events accumulate as coherent quantum phases rather than energy loss, the oscillation phase reads:

$$\phi_{\text{venting}} = p_{\text{vent}} \cdot \frac{\kappa^2 L}{\hbar c E_\nu}.$$

For $\phi_{\text{venting}} = \Delta m_{\text{atm}}^2 \cdot L / (4 \hbar c E_\nu)$ (standard phase, as in the equivalence section below), one requires:

$$p_{\text{vent}} = \frac{\Delta m_{\text{atm}}^2}{4 \kappa^2} = \frac{120 m_\nu^2}{4 \cdot 144 m_\nu^2} = \frac{120}{576} = \frac{5}{24}.$$

The factor 4 of the standard phase is essential here: without it one would obtain $p = 120/144 = 5/6$.

Algebraic derivation of $p = 5/24$ from the group structure [B]

The branching rule for restricting the 4D representation ρ_4 of A_5 to the subgroup A_4 (tetrahedral group, order 12) reads:

$$\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}.$$

This follows from the A_4 character table: the characters of ρ_4 on the A_4 classes (e , (123) , (132) , $(12)(34)$) are $(4, 1, 1, 0)$, and the scalar products with A_4 characters yield multiplicity 1 for both $\rho_1^{A_4}$ and $\rho_3^{A_4}$.

Consequence: A_4 leaves a one-dimensional sub-representation (the W-direction) fixed in ρ_4 . That is, A_4 is the *stabiliser* of the W-direction in A_5 .

In A_5 there are exactly

$$|A_5 : A_4| = \frac{60}{12} = 5$$

conjugates of A_4 – corresponding to the 5 tetrahedra embedded in the icosahedron. Each conjugate stabilises a different W-direction. The W-venting selects exactly one of these 5 directions (the physical 4th spatial direction), hence:

$$p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

Equivalence to the standard oscillation formula

The complete SICC venting formula reads:

$$\phi_{\text{venting}} = \frac{5}{24} \cdot \frac{\kappa^2 L}{\hbar c E_\nu} = \frac{5}{24} \cdot \frac{144 m_\nu^2 L}{\hbar c E_\nu} = \frac{120 m_\nu^2 \cdot L}{4 \hbar c E_\nu} = \frac{\Delta m_{\text{atm}}^2 \cdot L}{4 \hbar c E_\nu}.$$

This is *identical* to the standard oscillation formula. All parameters – κ , p_{vent} , Δm_{atm}^2 – derive from the Galois structure $GF(27)^*$ and the group structure $A_5 \supset A_4$. No free parameter, no DUNE calibration.

The complete derivation chain:

$$GF(27)^* \xrightarrow{[B]} m_\nu \xrightarrow{[K]} \kappa = 12 m_\nu \xrightarrow{[K]} \Delta m_{\text{atm}}^2 = 120 m_\nu^2 \xrightarrow{[B]} p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

Non-closure of the angular structure: $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ [B]

The SICC simulation quantises directions onto the 24 kissing vectors of D_4 ; the Frobenius structure supplies orbit phases $2\pi k/13$. The two angular lattices are coprime:

$$\gcd(13, 24) = 1, \quad \text{lcm}(13, 24) = 312.$$

The combined angular structure therefore closes only after 312 steps, not 24 – a discrete spiral. This is the non-closure structure of Doc. 193 in angular form: there at the level of mass products $r_i \cdot r_j$, here at the level of $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$.

Continuously, the spiral also appears across the beam spectrum: $\phi(E) \propto 1/E$ is not periodic in E . Over $[0.5, 5]$ GeV (DUNE) the phase winds through 1.155 turns with remainder $55.7^\circ \bmod 2\pi$ – the curve does not close. This is invisible in a 1D phase histogram; in a polar plot (radius = spectral weight, angle = $\phi \bmod 2\pi$) the spiral becomes visible.

Fractal correction: not applicable to ratios

The fractal correction $K_{\text{frak}} = 1 - 100\xi$ (Doc. 011/133) applies to absolute masses, not to dimensionless ratios (Doc. 306). $\Delta m_{\text{atm}}^2/m_\nu^2 = 120$ is a ratio; K_{frak} must not be applied here. As a check: applied, the deviation of Δm_{atm}^2 would rise from 1.0 % to 2.3 %; the effect on DUNE disappearance would be 0.2 percentage points – below detector resolution.

.5 Mixing Angles [K] / [S]

Two mixing angles can be read off from Orbit-2 elements with $< 6\%$ deviation; these are order-of-magnitude statements (see below) [S]:

$$\sin^2 \theta_{12} \approx \cos^2\left(\frac{2\pi \cdot 2}{13}\right) = 0.323 \quad (\text{measured: } 0.307, \text{ dev. } 5.1\%)$$

$$\sin^2 \theta_{23} \approx \cos^2\left(\frac{2\pi \cdot 5}{13}\right) = 0.560 \quad (\text{measured: } 0.545, \text{ dev. } 2.8\%)$$

The reactor angle θ_{13} has two independent Galois derivations [K]:

(a) Galois formula from Doc. 328 (stronger):

$$\theta_{13} = \arcsin((25\xi)^{1/3}), \quad 25\xi = \frac{1}{300} = \frac{1}{5 \cdot |A_5|},$$

$$\sin^2 \theta_{13} = \left(\frac{1}{300}\right)^{2/3} \approx 0.02230 \quad (\text{measured: } 0.0220, \text{ dev. } 1.4\%)$$

The factor $25 = 5^2$ comes from the Galois identity $43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$ (Doc. 338), where 5^2 is the square of the maximal A_5 representation dimension.

(b) Sum rule from R_ν (independent confirmation):

$$\sin^2 \theta_{13} = \frac{2}{3} R_\nu = \frac{2}{3} \cdot \frac{11}{360} = \frac{11}{540} \approx 0.02037 \quad (\text{dev. } 7.4\%) \quad [1]$$

The factor $2/3$ comes from the sum rule of the source; there is no Galois reading of this factor.

The same source independently gives:

$$\sin^2 \theta_{12} = \frac{1}{3 - 2R_\nu} \approx 0.340 \quad (\text{dev. from Galois value } 0.323: 5.4\%)$$

The CP-violation phase $\delta_{CP} \approx -\pi/2$ (experimental hint) corresponds to Orbit-3 element $k = 10$: $\arg(e^{2\pi i \cdot 10/13}) = -83.1^\circ$, deviation 6.9° **[S]**.

The $\cos^2(2\pi k/13)$ readings for θ_{12} and θ_{23} in this section are order-of-magnitude statements: the twelve values $\{\cos^2(2\pi k/13), 1 - \cos^2(2\pi k/13)\}$ are so dense that at 6 % tolerance almost any parameter between 0 and 1 finds a neighbour. For θ_{23} , Doc. 372 gives the sharper statement $\sin^2 \theta_{23} \in \{4/9, 5/9\}$ from the A_5 matrix-element spectrum (prediction $|\sin^2 \theta_{23} - 1/2| = 1/18$); $\cos^2(10\pi/13) = 0.560$ and $5/9 = 0.556$ are algebraically distinct (degree 6 versus rational) and only numerically close **[B]**. The derivation $\sin^2 \theta_{13} = (25\xi)^{2/3}$ above stands; it uses $|A_5|$ and ξ , not the $\cos^2$ reading. For θ_{12} only the order of magnitude remains **[S]**.

Literature connection: T_{13} as a flavour symmetry group

The Frobenius group $T_{13} = \mathbb{Z}_{13} \rtimes \mathbb{Z}_3$ is known in the literature as a candidate for the flavour symmetry of neutrinos **[2]**. It is isomorphic to the $\mathbb{Z}_{13}$ part of $GF(27)^*$ together with the Frobenius action, i.e. to the Frobenius structure on the orbits of $\mathbb{Z}_{13}$ used in this document. The literature treated T_{13} as an abstract symmetry (without physical derivation); FFGFT provides the algebraic origin: $T_{13} \cong (\mathbb{Z}_{13} \text{ part of } GF(27)^*) \rtimes \langle \phi \rangle$ with the Frobenius automorphism $\phi : x \mapsto x^3$; the full group is $GF(27)^* \rtimes \langle \phi \rangle \cong \mathbb{Z}_2 \times T_{13}$. ($|T_{13}| = 39$, the full group has order 78; the Frobenius fixes the $\mathbb{Z}_2$ part.)

.6 Orbit 3 as the Majorana Sector **[B]**

Orbit 3 $\{4, 10, 12\}$ is fully self-inverse in $\mathbb{Z}_{13}$:

$$4^{-1} \equiv 10, \quad 10^{-1} \equiv 4, \quad 12^{-1} \equiv 12 \pmod{13}.$$

The self-inverse nature means the associated particles are their own antiparticles – Majorana character. Additionally $4 + 10 + 12 = 26 = |GF(27)^*|$ **[B]**.

Doc. 343 (Theorem D''') shows the quadratic character is the general criterion: Orbit 1 is also self-inverse (Majorana, ν_1); Orbit 4 is not (Dirac, ν_3). A separate prediction of sterile neutrinos from the Galois algebra does not follow **[B]**.

.7 Neutrinoless Double Beta Decay **[K]**

With measured PMNS mixing angles and Galois masses:

$$m_{ee} = |c_{12}^2 c_{13}^2 m_1 + s_{12}^2 c_{13}^2 m_2| \approx 6.0 \text{ meV}$$

(Majorana phases zero; ν_3 Dirac $\Rightarrow$ no m_3 contribution, Doc. 343 **[B]**). The KamLAND-Zen bound $m_{ee} < 28\text{--}122 \text{ meV}$ (PRL 135 (2025) 262501; NME-dependent) is satisfied with large margin. Upper bound without ν_3 : $m_{ee} \leq m_{\nu_1} + m_{\nu_2} \approx 14.4 \text{ meV}$. Minimum (destructive interference): $m_{ee, \min} \approx 0.13 \text{ meV}$. With $s_{12}^2 = 0.307$, $s_{13}^2 = 0.022$, $m_1 = 4.54$, $m_2 = 9.81 \text{ meV}$ the two terms are 3.08 and 2.95 meV; their sum gives 6.0, their difference 0.13 meV. A term $s_{13}^2 m_3$ does not occur, since ν_3 is a Dirac neutrino.

.8 Conjugation Structure of the Orbits **[B]**

The four Frobenius orbits in $\mathbb{Z}_{13}$ form two conjugate $3/\bar{3}$ pairs under $x \mapsto -x \pmod{13}$, matching the conjugacy classes $C_{3_1}, C_{\bar{3}_1}, C_{3_2}, C_{\bar{3}_2}$ of T_{13} **[2]**:

$$-\text{Orbit}_1 = \text{Orbit}_3, \quad -\text{Orbit}_2 = \text{Orbit}_4.$$

Together with the inversion duality ($x \mapsto x^{-1}$, Doc. 339), the orbits carry a complete $\mathbb{Z}_2 \times \mathbb{Z}_3$ symmetry in $\mathbb{Z}_{13}$. Orbit 1 is also self-inverse under inversion ($3^{-1} = 9$, $9^{-1} = 3$, both in Orbit 1).

.9 Model Discrepancies for Context [S]

The A_4/TM_2 model [1] that provides the θ_{13} sum rule makes predictions that deviate from both the Galois picture and experiment:

- $\sin^2 \theta_{23} = 0.433\text{--}0.441$ (lower octant) – measured: $0.545\text{--}0.560$ (upper octant); Galois: 0.560 (upper octant).
- $\delta_{CP} \approx \pm 51^\circ$ – T2K hint: $\approx -90^\circ$; Galois candidate ($k = 10$ from Orbit 3, justification open [S]): $\approx -83^\circ$.
- $m_{ee} = 5.1\text{--}5.3$ meV (with Majorana phases) – Galois (zero phases): 6.0 meV.

For θ_{23} and δ_{CP} , the FFGFT Galois values agree better with experiment than the A_4 model. The θ_{13} sum rule is adopted; the remaining A_4 predictions are not.

.10 Falsifiable Predictions

1. **Neutrino masses (normal ordering):** $m_1 = 4.54$, $m_2 = 9.81$, $m_3 = 49.96$ meV, $\Sigma = 64.3$ meV. Testable: CMB-S4, Euclid (cosmology at 1 meV level). Independent convergence: Razzaghi et al. [1] extract from A_4 texture and experiment $m_2 = (9.21\text{--}9.92)$, $m_3 = (49.91\text{--}51.42)$, $\Sigma = (63.97\text{--}64.55)$ meV; our Galois values m_2 , m_3 and Σ lie within these ranges.
2. **Mass-difference ratio [K]:**

$$\frac{\Delta m_{\text{atm}}^2}{\Delta m_{\text{sol}}^2} = \frac{3(11^2 - 1)}{11} = 32.73 \quad (\text{measured: } 33.20).$$

Once m_ν is measured to 1 meV precision, the ratio $\Delta m^2/m_\nu^2$ is directly testable.

3. **ν_3 is a Dirac neutrino [B]:** No Majorana phase for ν_3 ; $m_{ee} \leq m_{\nu_1} + m_{\nu_2} \approx 14.4$ meV. Inverted hierarchy more strongly excluded. Testable: nEXO ($m_{\beta\beta} \approx 5\text{--}20$ meV, NME-dependent); KamLAND-Zen-800: $< 28\text{--}122$ meV.
4. **Venting rate and DUNE observable [K]:** The coherent SICC venting amplitude $p_{\text{vent}} = 5/24 \approx 20.8\%$ per coherence step is a model-internal prediction. The DUNE observable depends on the beam spectrum: monochromatic 2.5 GeV gives 99.8% ν_μ disappearance (first oscillation maximum, $\phi = 1.61$ rad); a Gaussian spectrum 2.5 ± 1.0 GeV clamped to $[0.5, 5]$ GeV gives an average of 76% (SICC V3: 76.16%). For a direct prediction the real LBNF flux table must be used as weight.

.11 Open Points

1. Connection of the discrete orbit phases $2\pi k/13$ to the full PMNS U -matrix [S].
2. Algebraic justification for why $k = 10$ from Orbit 3 is distinguished for δ_{CP} ; candidate $\arg(e^{2\pi i \cdot 10/13}) = -83.1^\circ$ (T2K hint $\approx -90^\circ$, dev. 6.9°) [S].

.12 Note on Authorship

This document was developed on 30 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits in $\mathbb{Z}_{13}$, branching rule $\rho_4 \downarrow_{A_4}$, character theory of A_5 and A_4 , numerical verification of the mass differences) were carried out by the assistant and verified by the author. The physical interpretation, the connection to the SICC framework, and all status markers (**[B]**, **[K]**, **[S]**) are the author's responsibility.

.13 Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

.14 Verification Scripts

pruef_340_neutrino_galois.py (13/13 assertions passed);
 pruef_340b_spirale_breitband.py (6/6 assertions passed: non-closure $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$, spiral, broadband DUNE, K_{frak}).

Declared 30 August 2026, supplemented 31 August 2026 (θ_{13} , T_{13} literature, non-closure, broadband)

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